

Cryptography Final Project

Object :

Capstone Project

Date:

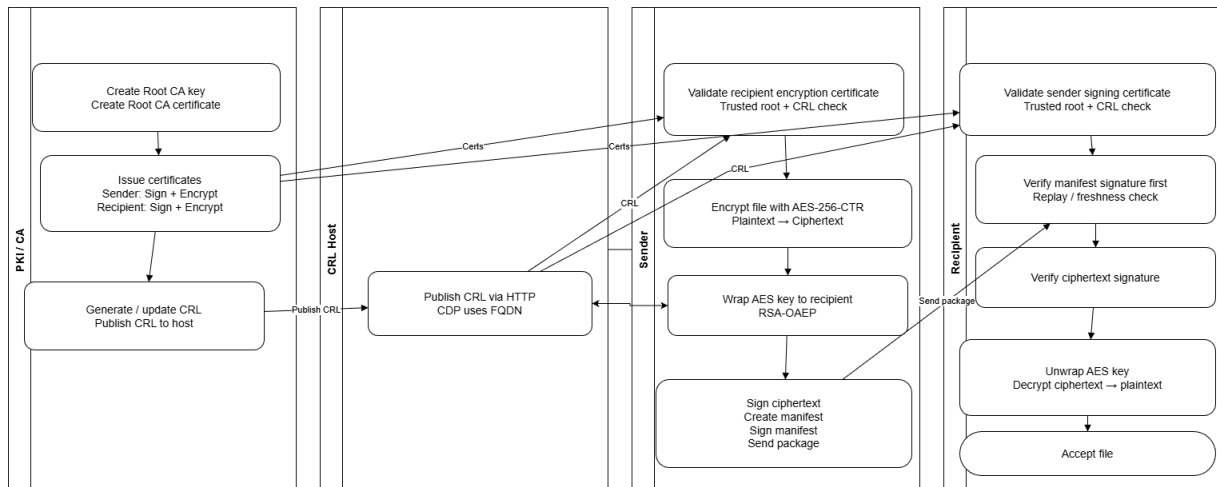
04/12/2025

Work Overflow:

CST8805 Team 14 – Secure File Exchange Workflow (Simple)

Workflow Overview

PKI issues separate signing and encryption certificates and publishes a CRL over HTTP using an FQDN. Sender validates the recipient encryption certificate, encrypts the file, wraps the AES key to the recipient, signs the encrypted file, and creates a signed manifest. Recipient validates the sender signing certificate, verifies the manifest first, checks replay/freshness, then unwraps the key, decrypts the file, and verifies the signature.



1. Root CA Setup

Machine 1: PKI

STEP 1: Part 1

- Create Root CA private key

Generate CA Private key:

```
D:\STUDIES\Crypto\CST8805\PROJECT2>openssl genpkey -algorithm RSA -pkeyopt rsa_keygen_bits:4096 -aes-256-cbc -out CA-Pri
v.key
```

Creating self-signed Root CA certificate:

```
D:\STUDIES\Crypto\CST8805\PROJECT2>>openssl req -x509 -new -key CA-Priv.key -days 365 -sha256 -out CARootCert.cer
Enter pass phrase for CA-Priv.key:
```

```
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:CA
State or Province Name (full name) [Some-State]:On
Locality Name (eg, city) []:Ottawa
Organization Name (eg, company) [Internet Widgits Pty Ltd]:team14
Organizational Unit Name (eg, section) []:team14
Common Name (e.g. server FQDN or YOUR name) []:team14
Email Address []:team14@team14.com
```

Generate Initial CRL:

```
D:\STUDIES\Crypto\CST8805\PROJECT2>openssl ca -config openssl.cnf -gencrl -out CAProject.crl.pem
```

Convert CRL to DER Format:

```
D:\STUDIES\Crypto\CST8805\PROJECT2>openssl crl -in CAProject.crl.pem -outform DER -out crlteam14.crl
```

```
D:\STUDIES\Crypto\CST8805\PROJECT2>dir crlteam14.crl
```

```
Volume in drive D is QUADRATSH
Volume Serial Number is 34C1-2060
```

Check CRL encoding:

```
D:\STUDIES\Crypto\CST8805\PROJECT2>openssl crl -inform DER -in crlteam14.crl -noout -text
Certificate Revocation List (CRL):
  Version 2 (0x1)
  Signature Algorithm: sha256WithRSAEncryption
  Issuer: C = CA, ST = ONTARIO, L = OTTAWA, O = TEAM14, OU = PKI, CN = TEAM14ROOT, emailAddress = PKI.TEAM14@TEAM14.CA
  Last Update: Dec 2 00:35:44 2025 GMT
  Next Update: Jan 1 00:35:44 2026 GMT
  CRL extensions:
    X509v3 CRL Number:
      1
  No Revoked Certificates.
  Signature Algorithm: sha256WithRSAEncryption
  Signature Value:
    22:6a:75:1a:1f:2a:82:56:6b:e6:62:13:bb:76:15:7a:36:1e:
    01:ad:4c:8d:28:ed:aa:83:24:5d:96:b1:11:a2:15:c4:56:df:
    35:d5:1e:90:cf:ac:15:32:51:01:f1:0c:eb:ca:5c:da:a9:ea:
    48:3c:9e:7a:d8:34:8f:aa:b7:5d:c6:bc:7c:ef:39:f7:5e:55:
    4f:05:1a:80:29:6e:56:62:31:ee:12:a5:08:0e:e5:7b:4a:a9:
    ed:ab:36:84:4e:19:95:36:60:b7:8a:1e:03:9b:c1:94:eb:dd:
    40:92:3e:61:09:99:2a:52:22:12:27:2e:22:6e:e8:e6:b1:ba:
    ff:1f:f5:9a:e0:e5:f9:46:7a:51:94:23:b5:84:5a:49:21:c4:
    7e:1f:08:2b:ff:a4:ac:63:ea:a6:2f:ae:b5:55:69:a9:5f:a9:
    77:28:5b:d1:59:f7:a9:46:c7:da:66:22:9e:49:6b:cb:74:41:
    ed:a2:f6:36:02:37:00:28:a0:be:50:26:7f:34:b3:85:c6:87:
    ce:fc:97:9a:d1:c2:f6:53:96:36:61:78:63:b2:12:f4:58:e6:
    ab:5f:72:b8:0f:2f:51:4c:84:d9:7e:f4:88:ac:c9:08:88:7f:
    9b:e5:1d:8f:42:b9:72:e0:14:1c:01:aa:9b:37:da:39:22:70:
    f9:1f:4d:f0:d3:4f:e3:7f:c8:ad:ba:3d:e9:5b:fb:e3:84:81:
    a5:9c:c6:ec:81:26:56:a7:16:66:ea:30:cc:3e:78:39:69:6f:
    43:05:d9:bd:bb:85:f6:93:cc:1f:60:cd:a1:a7:c4:4d:c0:c2:
    83:ef:57:2b:06:21:0b:e9:b1:72:3f:44:0d:82:54:15:20:a7:
    22:55:f6:83:1d:7d:7a:b4:61:10:1c:8a:99:71:06:51:bd:76:
    ac:32:a6:2e:39:86:98:6f:50:b7:2b:d8:f1:5d:cd:b5:18:7a:
    d0:44:ac:f1:eb:e2:c8:ae:f6:75:94:23:df:41:cc:d0:45:f5:
    7c:41:12:e6:9b:94:38:51:58:fb:9c:fd:69:29:90:1a:62:f3:
    46:8f:26:f0:b8:74:75:89:a8:4f:29:4a:a5:13:9e:4b:4f:a7:
    93:5e:48:32:33:5d:95:16:e5:a7:dd:d1:23:4d:de:1c:aa:63:
    77:2a:e1:7f:da:3b:dc:fc:d5:aa:30:0b:d3:23:b7:d4:c2:8b:
    80:87:93:af:1e:51:0f:9c:e0:84:55:d2:9c:42:16:91:4e:1b:
    05:21:70:dd:df:ce:79:27:9b:40:62:bb:2d:4a:71:0c:3a:5e:
    63:c0:c0:ef:fc:2f:c4:17:30:3a:ec:a3:33:c8:f3:c8:49:24:
    72:30:5d:bf:4c:d5:0d:0d
```

Publish DER CRL via menu

The CRL becomes publicly available

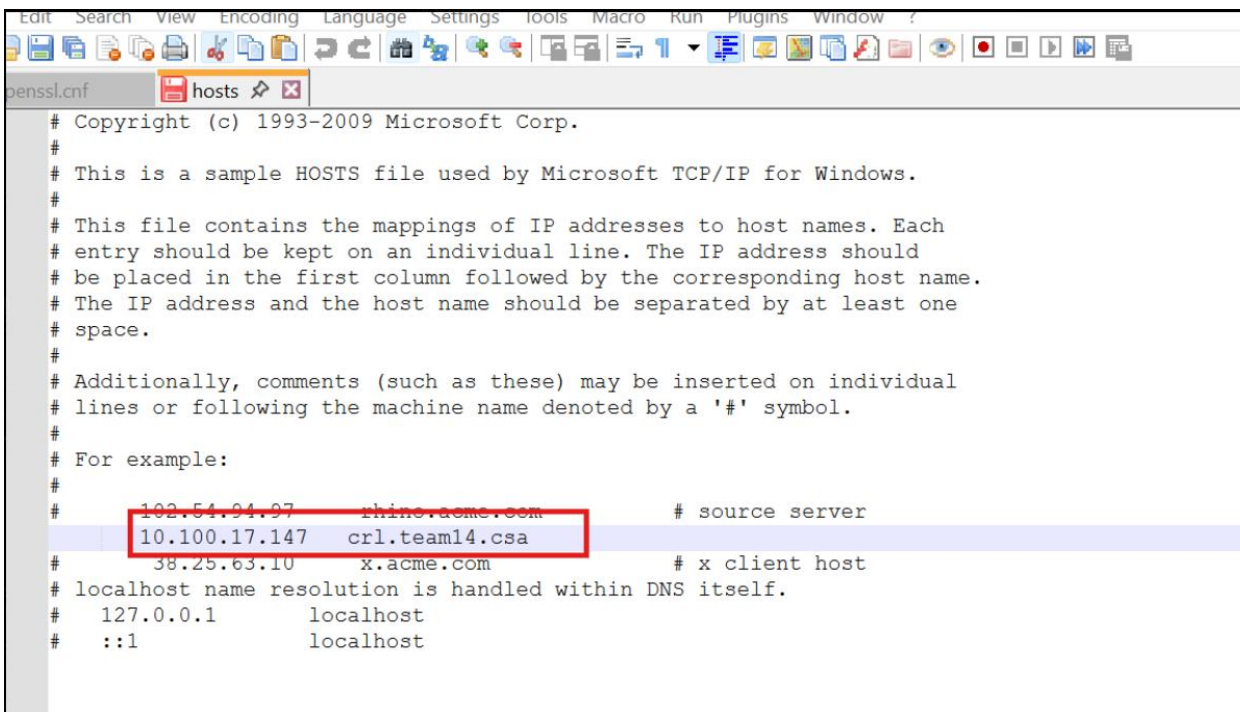
Project - Publish CRL

Your CRL has been published

- [Home](#)
- [Logout](#)

<http://crl.team14.csa:34535/crl/crlteam14.crl>

Edited the host on the sender machine



```
# Copyright (c) 1993-2009 Microsoft Corp.
#
# This is a sample HOSTS file used by Microsoft TCP/IP for Windows.
#
# This file contains the mappings of IP addresses to host names. Each
# entry should be kept on an individual line. The IP address should
# be placed in the first column followed by the corresponding host name.
# The IP address and the host name should be separated by at least one
# space.
#
# Additionally, comments (such as these) may be inserted on individual
# lines or following the machine name denoted by a '#' symbol.
#
# For example:
#
# 102.54.94.97 rhino.acme.com          # source server
# 10.100.17.147 crl.team14.csa
# 38.25.63.10 x.acme.com              # x client host
# localhost name resolution is handled within DNS itself.
# 127.0.0.1 localhost
# ::1 localhost
```

To ensure the Sender can resolve the CRL Distribution Point (CDP) FQDN, the local HOSTS file must be edited to include the entry 10.100.17.147 crl.team14.csa, enabling the required CRL check during certificate validation.

Machine2: Sender

The sender received the these from the root PKI

- CARootCert.cer
- send-Enc.cer
- crlteam14.der download from the shared link

Generate Signing & Encryption keys

Generate Sender encryption key:

```
C:\Users\Oussama\Desktop\PROJECT2SENDER>openssl genpkey -algorithm RSA -pkeyopt rsa_keygen_bits:2048 -aes-128-cbc -out S
ender-Enc.key
```

[illegible]

```
C:\Users\Oussama\Desktop\PROJECT2SENDER>openssl genpkey -algorithm RSA -pkeyopt rsa_keygen_bits:2048 -aes-128-cbc -out S  
ender-Sign.key
```

[illegible]

Create Sender signing CSR:

```
C:\Users\Oussama\Desktop\PROJECT2SENDER>openssl req -new -key Sender-Sign.key -out Sender-Sign.csr
Enter pass phrase for Sender-Sign.key:
```

```
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
```

```
Country Name (2 letter code) [AU]:CA
State or Province Name (full name) [Some-State]:ONTARIO
Locality Name (eg, city) []:OTTAWA
Organization Name (eg, company) [Internet Widgits Pty Ltd]:TEAM14
Organizational Unit Name (eg, section) []:TEAM14
Common Name (e.g. server FQDN or YOUR name) []:TEAM14
Email Address []:SENDER@TEAM14.CA
```

```
Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:cyberops
An optional company name []:cyberops
```

Create Sender encryption CSR:

```
C:\Users\Oussama\Desktop\PROJECT2SENDER>openssl req -new -key Sender-Enc.key -out Sender-Enc.csr
Enter pass phrase for Sender-Enc.key:
```

```
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
```

```
Country Name (2 letter code) [AU]:CA
State or Province Name (full name) [Some-State]:ONTARIO
Locality Name (eg, city) []:OTTAWA
Organization Name (eg, company) [Internet Widgits Pty Ltd]:TEAM14
Organizational Unit Name (eg, section) []:TEAM14
Common Name (e.g. server FQDN or YOUR name) []:TEAM14
Email Address []:SENDER@TEAM14.CA
```

```
Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:cyberops
An optional company name []:cyberops
```

Create extension files used when CA signs:

```
D:\STUDIES\Crypto\CST8805\PROJECT2>notepad Sender-Sign_ext.txt
D:\STUDIES\Crypto\CST8805\PROJECT2>notepad Sender-Sign_ext.txt
D:\STUDIES\Crypto\CST8805\PROJECT2>notepad Sender-Enc_ext.txt
D:\STUDIES\Crypto\CST8805\PROJECT2>notepad Recipient-Sign_ext.txt
D:\STUDIES\Crypto\CST8805\PROJECT2>notepad Recipient-Enc_ext.txt
```

Created the CSR for the recipient I then sent the Sender-Enc.csr and the Sender-Sign.csr and sent it over to the recipient for the next steps to be processed.

Machine3: recipient

Generate Recipient keys

[illegible]

Receiver Encryption Key:

Create Signature CSR:

```
C:\Users\Oussama\Documents\PROJECT2RECIPIENT>openssl req -new -key Recv-Sign.key -out Recv-Sign.csr
Enter pass phrase for Recv-Sign.key:
```

```
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
```

```
Country Name (2 letter code) [AU]:CA
State or Province Name (full name) [Some-State]:ONTARIO
Locality Name (eg, city) []:OTTAWA
Organization Name (eg, company) [Internet Widgits Pty Ltd]:RECIPIENT
Organizational Unit Name (eg, section) []:RECIPIENT
Common Name (e.g. server FQDN or YOUR name) []:RECIPIENT
Email Address []:RECIPIENT@TEAM14.CA
```

```
Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:cyberops
An optional company name []:cyberops
```

After creating the CSR I will now as the recipient sending the Recv-Sign.csr and Recv-Enc.csr files to the PKI Administrator.

Machine 1 PKI:

Sign Sender certificates:

Sender Signature Certificate:

```
D:\STUDIES\Crypto\CST8805\PROJECT2>openssl x509 -req -in Sender-Sign.csr -CA CARootCert.cer -CAkey CA-Priv.key -set_serial 0x101 -sha256 -days 90 -extfile Sender-Sign_ext.txt -out Sender-Sign.cer
Certificate request self-signature ok
subject=C = CA, ST = ONTARIO, L = OTTAWA, O = TEAM14, OU = TEAM14, CN = TEAM14, emailAddress = SENDER@TEAM14.CA
Enter pass phrase for CA-Priv.key:
```

Sender Encryption Certificate:

```
D:\STUDIES\Crypto\CST8805\PROJECT2>openssl x509 -req -in Sender-Enc.csr -CA CARootCert.cer -CAkey CA-Priv.key -set_serial 0x102 -sha256 -days 90 -extfile Sender-Enc_ext.txt -out Sender-Enc.cer
Certificate request self-signature ok
subject=C = CA, ST = ONTARIO, L = OTTAWA, O = TEAM14, OU = TEAM14, CN = TEAM14, emailAddress = SENDER@TEAM14.CA
Enter pass phrase for CA-Priv.key:
```

Sign Recipient certificates:

Recipient Signature Certificate:

```
D:\STUDIES\Crypto\CST8805\PROJECT2>openssl x509 -req -in Recv-Enc.csr -CA CARootCert.cer -CAkey CA-Priv.key -set_serial 0x202 -sha256 -days 90 -extfile Recipient-Enc_ext.txt -out Recv-Enc.cer
Certificate request self-signature ok
subject=C = CA, ST = ONTARIO, L = OTTAWA, O = RECIPIENT, OU = RECIPIENT, CN = ECIPIENT, emailAddress = RECIPIENT@TEAM14.CA
Enter pass phrase for CA-Priv.key:
```

Recipient Encryption Certificate:

```
D:\STUDIES\Crypto\CST8805\PROJECT2>openssl x509 -req -in Recv-Enc.csr -CA CARootCert.cer -CAkey CA-Priv.key -set_serial 0x202 -sha256 -days 90 -extfile Recipient-Enc_ext.txt -out Recv-Enc.cer
Certificate request self-signature ok
subject=C = CA, ST = ONTARIO, L = OTTAWA, O = RECIPIENT, OU = RECIPIENT, CN = ECIPIENT, emailAddress = RECIPIENT@TEAM14.CA
Enter pass phrase for CA-Priv.key:
```

Machine 2 Sender:

Verify cert chains and CRL

```
C:\Users\Oussama\Desktop\PROJECT2SENDER>openssl verify -trusted CARootCert.cer Sender-Sign.cer
Sender-Sign.cer: OK
```

```
C:\Users\Oussama\Desktop\PROJECT2SENDER>openssl verify -trusted CARootCert.cer Sender-Enc.cer
Sender-Enc.cer: OK
```

```
C:\Users\Oussama\Desktop\PROJECT2SENDER>openssl verify -trusted CARootCert.cer Recv-Enc.cer
Recv-Enc.cer: OK
```

```
C:\Users\Oussama\Desktop\PROJECT2SENDER>openssl verify -crl_check -CRLfile crlteam14.crl -trusted CARootCert.cer Recv-Enc.cer
Recv-Enc.cer: OK

C:\Users\Oussama\Desktop\PROJECT2SENDER>openssl verify -crl_check -CRLfile crlteam14.crl -trusted CARootCert.cer Sender-Sign.cer
Sender-Sign.cer: OK

C:\Users\Oussama\Desktop\PROJECT2SENDER>openssl crl -inform DER -in crlteam14.crl -noout -text
Certificate Revocation List (CRL):
  Version 2 (0x1)
  Signature Algorithm: sha256WithRSAEncryption
  Issuer: C = CA, ST = ONTARIO, L = OTTAWA, O = TEAM14, OU = PKI, CN = TEAM14ROOT, emailAddress = PKI.TEAM14@TEAM14.CA
  Last Update: Dec  2 00:35:44 2025 GMT
  Next Update: Jan  1 00:35:44 2026 GMT
  CRL extensions:
    X509v3 CRL Number:
      1
No Revoked Certificates.
  Signature Algorithm: sha256WithRSAEncryption
  Signature Value:
    22:6a:75:1a:1f:2a:82:56:6b:e6:62:13:bb:76:15:7a:36:1e:
    01:ad:4c:8d:28:ed:aa:83:24:5d:96:b1:11:a2:15:c4:56:df:
    35:d5:1e:90:cf:ac:15:32:51:01:f1:0c:eb:ca:5c:da:a9:ea:
    48:3c:9e:7a:d8:34:8f:aa:b7:5d:c6:bc:7c:ef:39:f7:5e:55:
    4f:05:1a:80:29:6e:56:62:31:ee:12:a5:08:0e:e5:7b:4a:a9:
    ed:ab:36:84:4e:19:95:36:60:b7:8a:1e:03:9b:c1:94:eb:dd:
    40:92:3e:61:09:99:2a:52:22:12:27:2e:22:6e:e8:e6:b1:ba:
    ff:1f:f5:9a:e0:e5:f9:46:7a:51:94:23:b5:84:5a:49:21:c4:
    7e:1f:08:2b:ff:a4:ac:63:ea:a6:2f:ae:b5:55:69:a9:5f:a9:
    77:28:5b:d1:59:f7:a9:46:c7:da:66:22:9e:49:6b:cb:74:41:
    ed:a2:f6:36:02:37:00:28:a0:be:50:26:7f:34:b3:85:c6:87:
    ce:fc:97:9a:d1:c2:f6:53:96:36:61:78:63:b2:12:f4:58:e6:
    ab:5f:72:b8:0f:2f:51:4c:84:d9:7e:f4:88:ac:c9:08:88:7f:
    9b:e5:1d:8f:42:b9:72:e0:14:1c:01:aa:9b:37:da:39:22:70:
    f9:1f:4d:f0:d3:4f:e3:7f:c8:ad:ba:3d:e9:5b:fb:e3:84:81:
    a5:9c:c6:ec:81:26:56:a7:16:66:ea:30:cc:3e:78:39:69:6f:
    43:05:d9:bd:bb:85:f6:93:cc:1f:60:cd:a1:a7:cd:4d:c0:c2:
    83:ef:57:2b:06:21:0b:e9:b1:72:3f:44:0d:82:54:15:20:a7:
    22:55:f6:03:1d:7d:7a:64:61:10:1c:8a:99:71:06:51:bd:76:
    ac:32:a6:2a:39:86:98:6f:50:b7:2b:d8:f1:5d:cd:b5:10:7a:
    d0:44:ac:f1:eb:e2:c8:ae:f6:75:94:23:df:41:cd:d0:45:f5:
    7c:41:12:e6:9b:94:30:51:58:b7:9c:fd:69:29:90:1a:62:f3:
    46:8f:26:f0:b8:74:75:89:a8:4f:29:4a:a5:13:9a:4b:4f:a7:
    93:5e:48:32:33:5d:95:16:e5:a7:dd:d1:23:4d:de:1c:aa:63:
    77:2a:e1:7f:da:3b:dc:fc:d5:aa:30:0b:d3:23:b7:d4:c2:8b:
    80:87:93:af:1e:51:0f:9c:e0:84:55:d2:9c:42:16:91:4e:1b:
    05:21:70:dd:df:ce:79:27:9b:40:62:bb:2d:4a:71:0c:3a:5e:
    63:c0:c0:ef:fc:2f:c4:17:30:3a:ec:a3:33:c8:f3:c8:49:24:
    72:30:5d:bf:4c:d5:0d:0d
```

Generate symmetric key and plaintext:

```
C:\Users\Oussama\Desktop\PROJECT2SENDER>openssl rand -hex 32 > symkey.hex
C:\Users\Oussama\Desktop\PROJECT2SENDER>type symkey.hex
2aafb039d495b88adda08cbccc81e89b2dea633c2046dbff4db8584e9a568034
C:\Users\Oussama\Desktop\PROJECT2SENDER>echo "THIS THE GROUP14 PROJECT TESTING THE ENCRYPTION FILE" > Plain.txt
C:\Users\Oussama\Desktop\PROJECT2SENDER>
```

Encrypt file with AES-256-CTR:

Wrap symmetric key with Recipient encryption public key:

```
C:\Users\Oussama\Desktop\PROJECT2SENDER>openssl x509 -pubkey -noout -in Recv-Enc.cer > Recv-Enc.pub
```

```
C:\Users\Oussama\Desktop\PROJECT2SENDER>openssl pkeyutil -encrypt -pubin -inkey Recv-Enc.pub -pkeyopt rsa_padding_mode:pkcs1 -in symkey.hex -out symkey.hex.rsa
```

Sign ciphertext:

```
C:\Users\Oussama\Desktop\PROJECT2SENDER>openssl dgst -sha3-512 -sign Sender-Sign.key -out Enc.bin.sig Enc.bin
Enter pass phrase for Sender-Sign.key:
```

Create manifest. json:

```
C:\Users\Oussama\Desktop\PROJECT2SENDER>openssl dgst -sha3-512 -sign Sender-Sign.key -out Enc.bin.sig Enc.bin
Enter pass phrase for Sender-Sign.key:
```

```
C:\Users\Oussama\Desktop\PROJECT2SENDER>notepad manifest.json
```

Sign Manifest:

```
C:\Users\Oussama\Desktop\PROJECT2SENDER>openssl dgst -sha3-512 -sign Sender-Sign.key -out manifest.sig manifest.json
Enter pass phrase for Sender-Sign.key:
```

Sender sends the files:

- Enc.bin.sig
- symkey.hex.rsa
- manifest.json

- manifest.sig

Machine 3 Recipient:

Verify CRL and certs:

```
C:\Users\Oussama\Documents\PROJECT2RECIPIENT>openssl verify -crl_check -CRLfile crlteam14.crl -trusted CARootCert.cer Recv-Sign.cer
Recv-Sign.cer: OK

C:\Users\Oussama\Documents\PROJECT2RECIPIENT>openssl verify -crl_check -CRLfile crlteam14.crl -trusted CARootCert.cer Recv-Enc.cer
Recv-Enc.cer: OK

C:\Users\Oussama\Documents\PROJECT2RECIPIENT>openssl verify -crl_check -CRLfile crlteam14.crl -trusted CARootCert.cer Sender-Sign.cer
Sender-Sign.cer: OK

C:\Users\Oussama\Documents\PROJECT2RECIPIENT>
```

Verify manifest signature:

```
C:\Users\Oussama\Documents\PROJECT2RECIPIENT>openssl dgst -sha3-512 -verify Sender-Sign.pub -signature manifest.sig manifest.json
Verified OK
```

Verify ciphertext signature:

```
C:\Users\Oussama\Documents\PROJECT2RECIPIENT>openssl dgst -sha3-512 -verify Sender-Sign.pub -signature Enc.bin.sig Enc.bin
Verified OK
```

Unwrap symmetric key using Recipient enc key:

```
C:\Users\Oussama\Documents\PROJECT2RECIPIENT>openssl pkeyutl -decrypt -inkey Recv-Enc.key -in symkey.hex.rsa -out symkey.hex
Enter pass phrase for Recv-Enc.key:
```

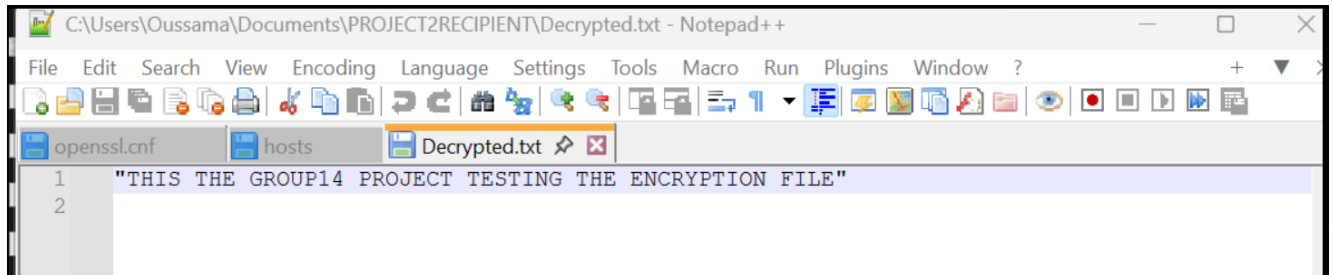
Extracting the key to decipher the cyphertext

```
C:\Users\Oussama\Documents\PROJECT2RECIPIENT>type symkey.hex
2aafb039d495b88adda08cbccc81e89b2dea633c2046dbff4db8584e9a568034
```

Decrypt ciphertext:

```
C:\Users\Oussama\Documents\PROJECT2RECIPIENT>openssl enc -d -aes-256-ctr -in Enc.bin -out Decrypted.txt -K a3261bc17b1e1b5ec51bafc0ffba5f5bd5be4eead5154fa7f805e20c670fb33a -iv 00000000000000000000000000000001
```

The content of the cyphertext after decrypting it



The recipient successfully validates the transmission by verifying the manifest and ciphertext signatures against the sender's trusted certificates checked via the CRL. Unwraps the symmetric session key using their private encryption key, and then finally uses that key to decrypt the ciphertext back into the original plaintext file, thereby achieving all security goals: authenticity, integrity, and confidentiality.